Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 100110 | *Subject: Engineering Physics* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / write the answer of the following (Any seven):*** **[2 x 7 = 14]**

(a) Which of the following principle explains that electrons do not exist inside the nucleus?
(i) Heisenberg's uncertainty principle (ii) Einstein's photo-electric effect
(iii) Huygen's principle (iv) All the above

(b) Which of the following is the characteristic of wave function?
(i) Continuous (ii) Single valued
(iii) Normalizable (iv) All of the above

(c) Which phenomenon explains the particle nature of light?
(i) Photoelectric effect (ii) Pair production
(iii) Blackbody radiation (iv) All the above

(d) Write down four applications of LASER in the fields of engineering.

(e) Define Fermi Energy.

(f) Write down time-dependent Schrödinger's wave equation in 1-dimension.

(g) Show graphically the Intensity distribution in Fraunhoffer diffraction due to a single slit.

(h) What is Lorentz force?

(i) Show that displacement vs. velocity graph of SHM is elliptical.

(j) Write down condition for the validity of ampere's circuital law?

Q.2 (a) Derive the differential equation for Damped Harmonic Oscillator. Give its solution. Differentiate between under damped, over damped and critically damped harmonic oscillation including graphical representation. **[10]**

(b) Explain Coriolis effect and its applications in weather system. **[4]**

Q.3 (a) Discuss the structure and working of Ruby LASER. Draw and explain its energy band diagram. **[10]**

(b) Write notes on population inversion in LASER. **[4]**

Q.4 (a) What is diffraction grating? Discuss the diffraction pattern due to a grating with intensities distribution curve. **[7]**

(b) Derive an expression for the fringe width in Young's double slit experiment **[7]**

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Q.5 (a) Write down Schrondiger equation for a particle in 1-D box given by – [10]

$V(x) = 0, \quad 0 < x < L$

$= \infty, \quad x < 0$ and $x > L$

Solve it to obtain energy eigen values. Show that energy eigen values are discrete.

(b) What is Compton effect? Explain how does Compton shift depend on angle of scattering? [4]

Q.6 (a) Discuss Kronig-Penney model for the motion of an electron in a periodic potential and the existence of allowed and forbidden energy bands. [10]

(b) Draw the total energy (E) vs. wave number (k) curve for an electron in a solid. [4]

Q.7 (a) State Biot-Savart's law. Apply it to derive an expression for magnetic flux density at the centre of a circular coil carrying current. [10]

(b) Distinguish among dia, para and ferro magnetic materials on the basis of orientation of atomic magnetic moments under the external magnetic fields. [4]

Q.8 (a) Define Poynting vector. Use Maxwell's equations to derive Poynting theorem. [10]

(b) State Faraday's laws and Lenz's law of electromagnetic induction. Write down the corresponding Maxwell's equation. [4]

Q.9 Write notes on ***any two*** of the following:- [7 x 2= 14]

(a) Einstein's photo-electric equation

(b) Polarization mechanisms in dielectrics

(c) Resonance

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